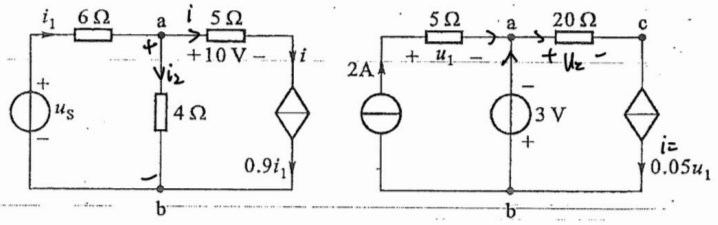
1-11 电路如题 1-11 图所示，试求：

(1) 图(a)中，i_1 与 u_{ab}。 (1) $i = \frac{10V}{5\Omega} = 2A = 0.9i_1$ $\Rightarrow i_1 = 2.222A$ $i_1 = i_2 + i \Rightarrow i_2 = 0.222A$ $U_{ab} = i_2 \times R = 0.889V$

(2) 图(b)中，u_{cb}。 (2) $U_1 = 5\Omega \times 2A = 10V$ $i = 0.05U_1 = 0.5A$ $U_2 = 0.5A \times 20\Omega = 10V$ $U_2 + U_{cb} + 3V = 0$

$\Rightarrow U_{cb} = -13V$



题 1-11 图

1-15 电路如题 1-15 图所示，试求每个元件发出或吸收的功率。

(a)

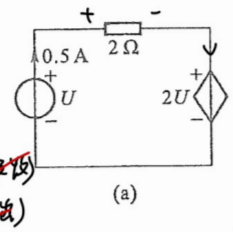
电阻：$P = I^2R = 0.5W$ (吸收)

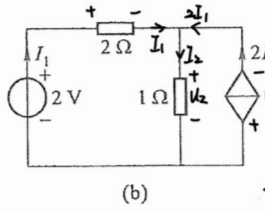
电压源：$-U + IR + 2U = 0$

$U = -1V$

$P_{发} = -1V \times 0.5A = -0.5W$ (吸收)

受控电压源：$P_{吸} = 0.5A \times 2U = -1W$ (发出)





(b) $I_1 + 2I_1 = I_2 = 3I_1$

$-2V + 2I_1 + U_3 = 0$

$U_2 = I_2R$

$\Rightarrow I_1 = 0.4A$ $I_2 = 1.2A$, $U_2 = 1.2V$

电压源：$P_{发} = 2V \times 0.4A = 0.8W$ (发出)

2Ω电阻：$P_{吸} = 0.4^2 \times 2 = 0.32W$ (吸收)

1Ω电阻：$P_{吸} = 1.2^2 \times 1 = 1.44W$ (吸收)

受控电流源：$U_3 + U_2 = 0$ $U_3 = -1.2V$

$P_{吸} = -1.2V \times 0.8A = -0.96W$ (发出)

题 1-15 图

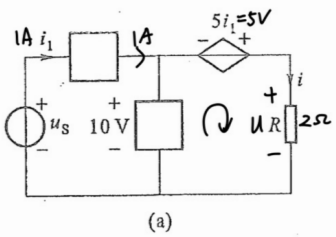
1-17 (1) 已知题 1-17 图(a)中，$R = 2\ \Omega$，$i_1 = 1\ A$，求电流 i。

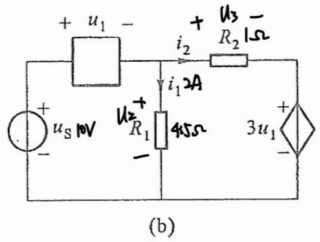
(2) 已知题 1-17 图(b)中，$u_S = 10\ V$，$i_1 = 2\ A$，$R_1 = 4.5\ \Omega$，$R_2 = 1\ \Omega$，求 i_2。

(1) $U - 10V - 5i_1 = 0$

$U = 15V$

$i = \frac{U}{R} = 7.5A$





(2) $U_2 = R_1 i_1 = 9V$

$U_1 + U_2 - U_S = 0$

$U_1 = 1V$

$U_3 + 3U_1 - U_2 = 0$

$U_3 = 6V$

$i_2 = \frac{U_3}{R_2} = 6A$

题 1-17 图